

# Laporan Kondisi Hardening Server

Windows Server 2022 — Referensi: CIS Microsoft Windows Server 2022 Benchmark v5.0.0

## Ringkasan Eksekutif

Total area hardening yang dikonfigurasi: 35 item. Dari jumlah tersebut, 26 item sudah sesuai dan tercakup dalam CIS Benchmark v5.0.0, 7 item merupakan konfigurasi beyond CIS (tidak ada di CIS namun valid sebagai defense-in-depth), dan 2 item berstatus partial atau telah dihapus dari versi CIS terbaru.

Secara keseluruhan, hardening yang telah dilakukan tidak melanggar rekomendasi CIS, bahkan melampaui standar minimum CIS di beberapa area seperti Exploit Guard, PowerShell Core, serta penonaktifan service yang tidak diperlukan (WebClient, Xbox).

## Keterangan Status

|                            |                                                                                  |
|----------------------------|----------------------------------------------------------------------------------|
| ✔ <b>Sesuai CIS</b>        | Konfigurasi ada dan sesuai dengan rekomendasi CIS Benchmark v5.0.0               |
| ❑ <b>Beyond CIS</b>        | Tidak ada di CIS v5.0.0 namun valid sebagai tambahan keamanan (defense-in-depth) |
| ❑ <b>Partial / Dihapus</b> | Konfigurasi partial atau telah di-remove dari CIS versi terbaru                  |

## Detail Kondisi Hardening

| Area Hardening                           | CIS Control | Level | Status              | Keterangan                                                 |
|------------------------------------------|-------------|-------|---------------------|------------------------------------------------------------|
| PowerShell Script Block Logging          | 18.10.88.1  | L2    | ✔ <b>Sesuai CIS</b> | Ada di CIS, sudah dikonfigurasi                            |
| PowerShell Transcription                 | 18.10.88.2  | L2    | ✔ <b>Sesuai CIS</b> | Ada di CIS, sudah dikonfigurasi                            |
| PowerShell Module Logging                | -           | -     | ❑ <b>Beyond CIS</b> | Tidak ada di CIS v5.0.0, tetapi best practice MITRE ATT&CK |
| PowerShell Core (PS 7+) Logging          | -           | -     | ❑ <b>Beyond CIS</b> | Tidak ada di CIS Server 2022 v5                            |
| SEHOP (Structured Exception Handling)    | 18.4.5      | L1    | ✔ <b>Sesuai CIS</b> | Enable SEHOP di MS Security Guide                          |
| UAC Restrictions (network logons)        | 18.4.1      | L1    | ✔ <b>Sesuai CIS</b> | Apply UAC restrictions to local accounts                   |
| LSA Protection (LSASS protected process) | 18.9.27.2   | L1/NG | ✔ <b>Sesuai CIS</b> | Configures LSASS to run as protected process               |

| Area Hardening                               | CIS Control   | Level | Status        | Keterangan                                                         |
|----------------------------------------------|---------------|-------|---------------|--------------------------------------------------------------------|
| Custom SSPs/APs loaded into LSASS            | 18.9.27.1     | L1    | ✔ Sesuai CIS  | Disabled - DC only                                                 |
| Do not store LAN Manager hash                | 2.3.11.5      | L1    | ✔ Sesuai CIS  | Credential storage protection                                      |
| Block Microsoft Accounts                     | -             | -     | ❑ Dihapus CIS | REMOVED dari CIS v5.0.0 (Ticket #23560)                            |
| NTLM: LAN Manager auth level (NTLMv2 only)   | 2.3.11.7      | L1    | ✔ Sesuai CIS  | Send NTLMv2 only, Refuse LM & NTLM                                 |
| NTLM: Min session security (client)          | 2.3.11.9      | L1    | ✔ Sesuai CIS  | Require NTLMv2 + 128-bit encryption                                |
| NTLM: Min session security (server)          | 2.3.11.10     | L1    | ✔ Sesuai CIS  | Require NTLMv2 + 128-bit encryption                                |
| NTLM: Audit Incoming Traffic                 | 2.3.11.11     | L1    | ✔ Sesuai CIS  | Enable auditing for all accounts                                   |
| NTLM: Audit Outgoing Traffic                 | 2.3.11.13     | L1    | ✔ Sesuai CIS  | Audit all or higher                                                |
| Network Access: No anonymous enum SAM        | 2.3.10.2      | L1    | ✔ Sesuai CIS  | Do not allow anonymous enumeration of SAM                          |
| Network Access: Restrict remote calls to SAM | 2.3.10.11     | L1    | ✔ Sesuai CIS  | Administrators: Remote Access                                      |
| Firewall Domain: State ON                    | 9.1.1         | L1    | ✔ Sesuai CIS  | Enable semua profile                                               |
| Firewall Domain: Log size 16384 KB+          | 9.1.5         | L1    | ✔ Sesuai CIS  | Log size sesuai rekomendasi                                        |
| Firewall Domain: Log dropped packets         | 9.1.6         | L1    | ✔ Sesuai CIS  | Yes                                                                |
| Firewall Domain: Log allowed connections     | 9.1.7         | L1    | ✔ Sesuai CIS  | Yes                                                                |
| Firewall Private & Public: State ON          | 9.2.1 / 9.3.1 | L1    | ✔ Sesuai CIS  | Enable semua profile                                               |
| Disable LLMNR                                | 18.6.4.x      | L1    | ✔ Sesuai CIS  | Turn off Multicast/LLMNR                                           |
| Turn off Microsoft P2P Networking            | 18.6.10.2     | L1    | ✔ Sesuai CIS  | Disabled                                                           |
| Disable Guest Logon                          | 2.3.1.1       | L1    | ✔ Sesuai CIS  | Guest account status Disabled                                      |
| WLAN Auto-connect                            | -             | -     | ❑ Partial CIS | CIS ada sleep/standby restriction, bukan auto-connect eksplisit    |
| Disable WebClient service                    | -             | -     | ❑ Beyond CIS  | Tidak ada di CIS Server 2022, valid attack surface reduction       |
| Disable Xbox services                        | -             | -     | ❑ Beyond CIS  | Tidak relevan di CIS Server benchmark, valid untuk hardening       |
| SEHOP                                        | 18.4.5        | L1    | ✔ Sesuai CIS  | Sudah tercakup di MS Security Guide                                |
| CFG (Control Flow Guard)                     | -             | -     | ❑ Beyond CIS  | Tidak di-cover CIS via GPO, dikonfigurasi via WDEG/Security Center |

| Area Hardening                  | CIS Control | Level | Status              | Keterangan                                                 |
|---------------------------------|-------------|-------|---------------------|------------------------------------------------------------|
| DEP (Data Execution Prevention) | -           | -     | ☐ <b>Beyond CIS</b> | Tidak di-cover CIS via GPO                                 |
| ASLR / High Entropy ASLR        | -           | -     | ☐ <b>Beyond CIS</b> | Tidak di-cover CIS via GPO, bagian dari Exploit Protection |
| Configure Redirection Guard     | 18.7.2      | L1    | ✔ <b>Sesuai CIS</b> | Enabled: Redirection Guard Enabled                         |
| Point and Print: new connection | 18.7.11     | L1    | ✔ <b>Sesuai CIS</b> | Show warning and elevation prompt                          |
| Point and Print: update driver  | 18.7.12     | L1    | ✔ <b>Sesuai CIS</b> | Show warning and elevation prompt                          |